

Day 3: Introduction to IP Addressing Theory (1 hour): IPv4 and IPv6: Differences, structure, and importance. Subnetting basics: CIDR, subnet masks. Practical (1 hour): Assign static IPs to devices and verify connectivity. Task: Solve subnetting exercises and calculate valid subnets.

Task 1.

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in decimal format, e.g. 192.168.1.10.	Written in hexadecimal format, e.g. 2001:db8::1.
Provides about 4.3 billion addresses.	Provides a vastly larger number of addresses, about 3.4×10^{38} .
Uses ARP to find the MAC address of a device.	Uses Neighbor Discovery Protocol (NDP) instead of ARP.
Broadcast communication is supported.	There is no broadcast; multicast and anycast are used.
Common subnet masks include /24, /25, /26, etc.	Common LAN subnet size is /64.

Task 2.

Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Network: 192.168.10.0/24

- Subnet mask: 255.255.255.0
- Default gateway: 192.168.10.1
- Network address: 192.168.10.0

- Broadcast address: 192.168.10.255
- Usable range: 192.168.10.1 – 192.168.10.254

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Task 3.

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

A /27 mask is:

255.255.255.224

Number of subnets

Assuming the original network is 192.168.1.0/24:

- Borrowed bits = $27 - 24 = 3$
- Number of subnets = $2^3 = 8$

Hosts per subnet

- Host bits = $32 - 27 = 5$
- Total addresses = $2^5 = 32$
- Usable hosts = $32 - 2 = 30$

So:

8 valid subnets, with 30 usable hosts per subnet.

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Task 4.

You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

The first address after the network address and the address before broadcast are usable:

Valid host range = 10.0.5.1 – 10.0.5.254

IP Address	10.0.5.17
CIDR	/24
Subnet Mask	255.255.255.0
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254
Usable Hosts	254

Task 5.

Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	0000101000000000000000101000100
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

Network Class ☐ Any ☐ A ☐ B ☒ C

Subnet

IP Address

The subnet calculator output matched my manual calculation: the network address is 10.0.5.0, broadcast address is 10.0.5.255, and the valid host range is 10.0.5.1–10.0.5.254.